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ON L^p–L^q TRACE INEQUALITIES

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Abstract. We give necessary and sufficient conditions in order that inequalities of the type

$$\|T_K f\|_{L^q(d\mu)} \le C\,\|f\|_{L^p(d\sigma)}, \qquad f \in L^p(d\sigma),$$

hold for a class of integral operators $T_K f(x) = \int_{R^n} K(x,y)\, f(y)\, d\sigma(y)$ with nonnegative kernels, and measures $d\mu$ and $d\sigma$ on $\mathbf{R}^n$, in the case where $p > q > 0$ and $p > 1$.

An important model is provided by the dyadic integral operator with kernel $K_{\mathcal{D}}(x,y) \sum_{Q\in\mathcal{D}} K(Q)\chi_Q(x)\,\chi_Q(y)$, where $\mathcal{D} = \{Q\}$ is the family of all dyadic cubes in $\mathbf{R}^n$, and $K(Q)$ are arbitrary nonnegative constants associated with $Q \in \mathcal{D}$.

The corresponding continuous versions are deduced from their dyadic counterparts. In particular, we show that, for the convolution operator $T_k f = k \star f$ with positive radially decreasing kernel $k(|x-y|)$, the trace inequality

$$\|T_k f\|_{L^q(d\mu)} \le C\,\|f\|_{L^p(dx)}, \qquad f \in L^p(dx),$$

holds if and only if $\mathcal{W}_k[\mu] \in L^s(d\mu)$, where $s = \frac{q(p-1)}{p-q}$. Here $\mathcal{W}_k[\mu]$ is a nonlinear Wolff potential defined by $\mathcal{W}_k[\mu](x) = \int_0^{+\infty} k(r)\overline{k}(r)^{\frac{1}{p-1}}\,\mu(B(x,r))^{\frac{1}{p-1}}\, r^{n-1}\, dr$, and $\overline{k}(r) = \frac{1}{r^n}\int_0^r k(t)\, t^{n-1}\, dt$. Analogous inequalities for $1 \le q < p$ were characterized earlier by the authors using a different method which is not applicable when $q < 1$.

1. Introduction

We consider inequalities of the type

$$\|T_K f\|_{L^q(d\mu)} \le C\,\|f\|_{L^p(d\sigma)}, \qquad f \in L^p(d\sigma),$$

where $d\sigma$ and $d\mu$ are nonnegative Borel measures on $\mathbf{R}^n$, and

$$T_K[f d\sigma](x) = \int_{\mathbf{R}^n} K(x,y)\, f(y)\, d\sigma(y), \qquad x \in \mathbf{R}^n,$$

is an integral operator with nonnegative kernel $K(x,y)$. Our goal is to obtain explicit necessary and sufficient conditions on K, μ, and σ which ensure that such inequalities hold for a broad class of dyadic and radially nonincreasing kernels in the case $0 < q < p < \infty$ and $p > 1$.

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We observe that such trace inequalities for $q = 1$ are related to the so-called Wolff inequality and nonlinear Wolff potentials which appeared originally in [**HW**] in connection with deep approximation properties of Sobolev spaces. (See [**AH**] where nonlinear potential theory and its applications are presented.) Generalizations of Wolff's inequality and appropriate versions of Wolff's potentials served as our main tools in earlier work [**COV1**], [**COV2**] where trace inequalities were characterized in the "upper triangle case" $1 \le q < p < \infty$. However, methods used there where based on duality arguments which are not applicable if $0 < q < 1$.

In this paper, we are mostly interested in the difficult case where $0 < q < 1$ and $p > 1$. Notice that our approach also works for $1 \le q < p < \infty$. We employ dyadic models, two weight Carleson measure theorems, and certain imbedding and interpolation theorems for discrete Littlewood–Paley spaces. Some of these techniques developed in the framework of trace inequalities turned out to be useful in applications to quasilinear and fully nonlinear PDE (see [**PhV**]).

An important special case is given by dyadic integral operators $T_{K_{\mathcal{D}}}$ and the corresponding integral inequalities from which continuous versions will be deduced. Let $\mathcal{D} = \{Q\}$ be the family of all dyadic cells $Q = 2^i(k + [0, 1)^n)$, $i \in \mathbf{Z}$, $k \in \mathbf{Z}^n$, and let r_Q denote the side length of Q. For $K : \mathcal{D} \to \mathbf{R}^+$, the kernel $K_{\mathcal{D}}(x, y)$ on $\mathbf{R}^n \times \mathbf{R}^n$ is defined by

$$K_{\mathcal{D}}(x, y) = \sum_{Q \in \mathcal{D}} K(Q)\, \chi_Q(x)\, \chi_Q(y),$$

where χ_Q is the characteristic function of $Q \in \mathcal{D}$.

Let ν be a nonnegative Borel measure on $\mathbf{R}^n$, and let $f \in L^1_{\mathrm{loc}}(d\nu)$. We define the dyadic integral operator:

$$T_{K_{\mathcal{D}}}[f\, d\nu](x) = \int_{\mathbf{R}^n} K_{\mathcal{D}}(x, y) f(y)\, d\nu(y) = \sum_{Q \in \mathcal{D}} K(Q) \chi_Q(x) \int_Q f\, d\nu.$$

We denote by $\overline{K}(Q)(x)$ the function

$$\overline{K}(Q)(x) = \frac{1}{\sigma(Q)} \sum_{Q' \subset Q} K(Q') \sigma(Q') \chi_{Q'}(x),$$

supported on Q, where σ is a nonnegative Borel measure on $\mathbf{R}^n$, and $\overline{K}(Q)(x) = 0$ if $\sigma(Q) = 0$.

For $x \in \mathbf{R}^n$, the generalized Wolff potential was defined in [**COV2**] as

$$\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\nu](x) = \sum_{Q \in \mathcal{D}} K(Q) \left[\int_Q \overline{K}(Q)(y)\, d\nu(y) \right]^{p'-1} \sigma(Q)\, \chi_Q(x). \tag{1.1}$$

Let us assume that the pair (K, σ) satisfies the dyadic logarithmic bounded oscillation condition (DLBO):

$$\sup_{x \in Q} \overline{K}(Q)(x) \le A \inf_{x \in Q} \overline{K}(Q)(x), \tag{1.2}$$

where A does not depend on $Q \in \mathcal{D}$. If K is a radially nonincreasing kernel and $d\sigma = dx$, or if $K(Q) = r_Q^{n-\alpha}$, $0 < \alpha < n$, and σ satisfies an adequate dyadic reverse doubling condition, then the pair (K, σ) satisfies condition (DLBO) (see [**COV2**]).

For $(K, \sigma) \in$ (DLBO), we set $\overline{K}(Q) = \inf_{x \in Q} \overline{K}(Q)(x)$, $Q \in \mathcal{D}$. In this case the definition of Wolff's potential can be simplified:

$$\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\nu](x) = \sum_{Q \in \mathcal{D}} K(Q)\, [\overline{K}(Q)]^{p'-1}\, [\nu(Q)]^{p'-1} \sigma(Q)\, \chi_Q(x). \tag{1.3}$$

The following theorem is proved in Section 2.

Theorem A. *Let $K : \mathcal{D} \to \mathbf{R}^+$, $0 < q < p < +\infty$, $1 < p < +\infty$. Let μ and σ be nonnegative Borel measures on $\mathbf{R}^n$. Suppose that $(K, \sigma) \in$ (DLBO). Then the trace inequality*

$$\int_{\mathbf{R}^n} |\, T_{K_{\mathcal{D}}}[f d\sigma]\, |^q \, d\mu \le C\, \|f\|^q_{L^p(d\sigma)}, \qquad f \in L^p(d\sigma), \tag{1.4}$$

holds if and only if $\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\mu] \in L^{\frac{q(p-1)}{p-q}}(d\mu)$.

In Section 3, we treat continuous trace inequalities. We state a version of Theorem A for integral operators with radial kernels,

$$T_k[f\, d\sigma](x) = \int_{\mathbf{R}^n} k(|x-y|)\, f(y)\, d\sigma(y).$$

Here $k = k(r)$, $r > 0$, is an arbitrary lower semicontinuous nonincreasing positive function. The corresponding nonlinear potential is defined by

$$\mathcal{W}_{k,\sigma}[\mu](x) = \int_0^{+\infty} k(r)\, \sigma(B(x,r)) \, \left(\int_{B(x,r)} \overline{k}(r)(y)\, d\mu(y) \right)^{p'-1} \frac{dr}{r},$$

where

$$\overline{k}(r)(x) = \frac{1}{\sigma(B(x,r))} \int_0^r k(s)\, \sigma(B(x,s))\, \frac{ds}{s},$$

for $x \in \mathbf{R}^n$, $r > 0$.

Theorem B. *Let $0 < q < p < +\infty$, $1 < p < \infty$, and let μ and σ be nonnegative Borel measures on $\mathbf{R}^n$. Assume that σ satisfies a doubling condition, and the pair (k, σ) has the following logarithmic bounded oscillation property* (LBO)*:*

$$\sup_{y \in B(x,r)} \overline{k}(r)(y) \le A \inf_{y \in B(x,r)} \overline{k}(r)(y), \tag{1.5}$$

where A does not depend on $x \in \mathbf{R}^n$, $r > 0$. Then the trace inequality

$$\int_{\mathbf{R}^n} |\, T_k[f d\sigma]\, |^q \, d\mu \le C\, \|f\|^q_{L^p(d\sigma)}, \qquad f \in L^p(d\sigma), \tag{1.6}$$

holds if and only if $\mathcal{W}_{k,\sigma}[\mu] \in L^{\frac{q(p-1)}{p-q}}(d\mu)$.

The (LBO) property is satisfied by all radially nondecreasing kernels in the case $d\sigma = dx$, or by Riesz kernels $k(x) = |x|^{\alpha-n}$, $0 < \alpha < n$, if σ satisfies an adequate reverse doubling condition (see [**COV2**]). In particular, the following corollary holds for convolution operators $T_k[f] = k \star f$ and $d\sigma = dx$.

Corollary C. *Let $0 < q < p < +\infty$ and $1 < p < \infty$. Let μ be a nonnegative Borel measure on $\mathbf{R}^n$. Suppose $k = k(|x-y|)$, where $k(r)$ is a lower semicontinuous nonincreasing positive function on $\mathbf{R}_+$, and $\overline{k}(r) = \frac{1}{r^n}\int_0^r k(t)\, t^{n-1}\, dt$. Then the trace inequality*

$$||k \star f||_{L^q(d\mu)} \le C\, \|f\|_{L^p(dx)}, \qquad f \in L^p(dx), \tag{1.7}$$

holds if and only if

$$\mathcal{W}_k[\mu](x) = \int_0^{+\infty} k(r)\, \overline{k}(r)^{\frac{1}{p-1}} \mu(B(x,r))^{\frac{1}{p-1}}\, r^{n-1}\, dr \in L^{\frac{q(p-1)}{p-q}}(\mu). \tag{1.8}$$

Remark 1. For Riesz kernels, a proof of Corollary C was sketched previously in [**V2**]. However, some technical details related to passing from discrete to continuous versions using shifts of the dyadic lattice are given below for the first time.

Remark 2. A more complicated characterization of (1.7) for Bessel kernels was given earlier in [**MN**] in terms of a certain capacity condition.

2. Discrete trace inequalities

In this section we will consider discrete trace inequalities. Recall that if $\mathcal{D} = \{Q\}$ denote the family of all dyadic cells, and $K : \mathcal{D} \to \mathbf{R}^+$, we have defined the kernel $K_{\mathcal{D}}(x,y)$ on $\mathbf{R}^n \times \mathbf{R}^n$ by

$$K_{\mathcal{D}}(x,y) = \sum_{Q\in\mathcal{D}} K(Q)\, \chi_Q(x)\, \chi_Q(y).$$

If ν is a nonnegative Borel measure on $\mathbf{R}^n$, and $f \in L^1_{\rm loc}(d\nu)$, the dyadic integral operator is given by

$$T_{K_{\mathcal{D}}}[f\, d\nu](x) = \int_{\mathbf{R}^n} K_{\mathcal{D}}(x,y) f(y)\, d\nu(y) = \sum_{Q\in\mathcal{D}} K(Q)\chi_Q(x) \int_Q f\, d\nu.$$

If $f \equiv 1$, we write $T_{K_{\mathcal{D}}}[\nu]$ in place of $T_{K_{\mathcal{D}}}[f\, d\nu]$.

Let σ be another nonnegative Borel measure on $\mathbf{R}^n$. As was already mentioned in the introduction, we will assume that the pair (K,σ) satisfies the dyadic logarithmic bounded oscillation condition (DLBO):

$$\sup_{x\in Q} \overline{K}(Q)(x) \le A \inf_{x\in Q} \overline{K}(Q)(x), \tag{2.1}$$

where A does not depend on $Q \in \mathcal{D}$, and

$$\overline{K}(Q)(x) = \frac{1}{\sigma(Q)} \sum_{Q'\subset Q} K(Q')\sigma(Q')\chi_{Q'}(x).$$

For $(K,\sigma) \in$ (DLBO), we set $\overline{K}(Q) = \inf_{x\in Q} \overline{K}(Q)(x)$, $Q \in \mathcal{D}$, where $\overline{K}(Q) = 0$ if $\sigma(Q) = 0$. Then the generalized Wolff potential introduced in [**COV2**] can be defined alternatively in an equivalent way by:

$$\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\nu](x) = \sum_{Q\in\mathcal{D}} K(Q)\,[\overline{K}(Q)]^{p'-1}\,[\nu(Q)]^{p'-1}\sigma(Q)\,\chi_Q(x). \tag{2.2}$$

The generalized Wolff inequality proved in [**COV2**] (Theorem A) yields:

$$C_1 \int_{\mathbf{R}^n} \mathcal{W}^{\mathcal{D}}_{K,\sigma}[\nu]\,d\nu \le \int_{\mathbf{R}^n} (T_{K_{\mathcal{D}}}[\nu])^{p'}\,d\sigma \le C_2 \int_{\mathbf{R}^n} \mathcal{W}^{\mathcal{D}}_{K,\sigma}[\nu]\,d\nu, \tag{2.3}$$

i.e., the energy of the measure ν given by $\int_{\mathbf{R}^n} (T_{K_{\mathcal{D}}}[\nu])^{p'}\,d\sigma$ is equivalent, under the (DLBO) assumption, to

$$\int_{\mathbf{R}^n} \mathcal{W}^{\mathcal{D}}_{K,\sigma}[\nu]\,d\nu = \sum_{Q\in\mathcal{D}} K(Q)\,\sigma(Q)[\overline{K}(Q)]^{p'-1}[\nu(Q)]^{p'}.$$

Theorem 2.1. *Let $K : \mathcal{D} \to \mathbf{R}^+$, $0 < q < p < +\infty$, and $1 < p < +\infty$. Let μ and σ be nonnegative Borel measures on $\mathbf{R}^n$. Suppose that $(K,\sigma) \in$ (DLBO). Then there exists a constant $C > 0$ such that the trace inequality*

$$\int_{\mathbf{R}^n} |\,T_{K_{\mathcal{D}}}[f\,d\sigma]\,|^q\,d\mu \le C\,\|f\|^q_{L^p(d\sigma)}, \qquad f \in L^p(d\sigma), \tag{2.4}$$

holds if and only if

$$\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\mu] \in L^{\frac{q(p-1)}{p-q}}(d\mu). \tag{2.5}$$

Moreover, the least constant C in (2.4) *is equivalent to $||\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\mu]||^{\frac{q}{p'}}_{L^{\frac{q(p-1)}{p-q}}(d\mu)}$ with constants of equivalence that depend only on p, q, and A.*

Remark. Note that the exponent $\frac{q(p-1)}{p-q} < 1$ in (2.5) if $0 < q < 1$. In the case $q = 1$, Theorem 2.1 is equivalent by duality to Wolff's inequality (2.3).

We first give a characterization of the trace inequality in the diagonal case $q = p$ in terms of discrete Carleson measures. It is deduced from the generalized Wolff inequality and the dyadic Carleson measure theorem. For Riesz potentials, this characterization was obtained earlier in [**V2**].

Lemma 2.2. *Let $K : \mathcal{D} \to \mathbf{R}^+$. Let $1 < p < +\infty$, and let μ and σ be nonnegative Borel measures on $\mathbf{R}^n$. Suppose that $(K,\sigma) \in$ (DLBO). Then the trace inequality* (2.4) *holds in the case $q = p$ if and only if there exists a constant B such that, for every dyadic cube $P \in \mathcal{D}$,*

$$\sum_{Q\subset P} K(Q)\,[\overline{K}(Q)]^{p'-1}\,[\mu(Q)]^{p'}\,\sigma(Q) \le B\,\mu(P). \tag{2.6}$$

Proof. By duality, (2.4) holds with $q = p$ if and only if

$$\int_{\mathbf{R}^n} |\,T_{K_{\mathcal{D}}}[g\,d\mu]\,|^{p'}\,(x)\,d\sigma(x) \le C\,\|g\|^{p'}_{L^{p'}(d\mu)}, \qquad g \in L^{p'}(d\mu). \tag{2.7}$$

Without loss of generality we may assume that $g \geq 0$. By the generalized Wolff inequality, the left-hand side is equivalent to

$$\sum_{Q\in\mathbf{D}} K(Q)\,[\overline{K}(Q)]^{p'-1}[\mu(Q)]^{p'}\sigma(Q)\,\left(\frac{1}{\mu(Q)}\int_Q g\,d\mu\right)^{p'}. \tag{2.8}$$

Applying the dyadic Carleson measure theorem in $L^{p'}(d\mu)$ (see, e.g., [**NTV**] or [**V1**], Sec. 3), we conclude that (2.7) holds if and only if (2.6) is valid. □

Lemma 2.3. *Let $1 < p < +\infty$. Let μ and σ be nonnegative Borel measures on $\mathbf{R}^n$. Suppose that $(K,\sigma) \in$ (DLBO) with constant A in (2.1). Define*

$$d\mu_1 = \frac{d\mu}{\left(\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\mu]\right)^{p-1}}.$$

Then

$$\int_{\mathbf{R}^n} |T_{K_{\mathcal{D}}}[f\,d\sigma]|^p\; d\mu_1 \leq C\,\|f\|^p_{L^p(d\sigma)}, \qquad f \in L^p(d\sigma), \tag{2.9}$$

where C depends only on p and A.

Proof of Lemma 2.3. By Lemma 2.2, the preceding inequality holds if and only if (2.6) is valid with μ_1 in place of μ. By Hölder's inequality,

$$[\mu_1(Q)]^{p'} \leq [\mu(Q)]^{p'-1}\,\int_Q \frac{d\mu}{\left(\mathcal{W}^{\mathcal{D}}_{K,\sigma}[\mu]\right)^{p}}.$$

Then by the preceding inequality, for any dyadic cube $P \in \mathcal{D}$,

$$\begin{aligned}
&\sum_{Q\subset P} K(Q)\,[\overline{K}(Q)]^{p'-1}\,[\mu_1(Q)]^{p'}\,\sigma(Q)\\
&\leq \sum_{Q\subset P} K(Q)\,[\overline{K}(Q)]^{p'-1}\,[\mu(Q)]^{p'-1}\,\sigma(Q)\\
&\times \int_Q \frac{d\mu_1(y)}{\sum_{Q'\in\mathcal{D}} K(Q')\,[\overline{K}(Q')]^{p'-1}\,[\mu(Q')]^{p'-1}\,\sigma(Q')\,\chi_{Q'}(y)}\\
&\leq \mu_1(P).
\end{aligned}$$

This proves (2.6) with μ_1 in place of μ, and $B = 1$. □

We now prove Theorem 2.1.

Proof of Theorem 2.1. We first prove that (2.5) is sufficient in order that (2.4) hold. Hölder's inequality with exponent $\frac{p}{q} > 1$, together with Lemma 2.2, yield

$$\begin{aligned}
&\left(\int_{\mathbf{R}^n} |T_{K_{\mathcal{D}}}[f\,d\sigma]|^q \, d\mu\right)^{\frac{1}{q}} \\
&\le \left(\int_{\mathbf{R}^n} |T_{K_{\mathcal{D}}}[f\,d\sigma]|^p \, \frac{d\mu}{\mathcal{W}^d_{K,\sigma}[\mu]^{p-1}}\right)^{\frac{1}{p}} \left(\int_{\mathbf{R}^n} \mathcal{W}^d_{K,\sigma}[\mu]^{\frac{q(p-1)}{p-q}} \, d\mu\right)^{\frac{p-q}{qp}} \\
&\le C \left(\int_{\mathbf{R}^n} |T_{K_{\mathcal{D}}}[f\,d\sigma]|^p \, d\mu_1\right)^{\frac{1}{p}} \le C \, \|f\|_{L^p(d\sigma)}.
\end{aligned}$$

This proves (2.4).

We now prove the necessity of (2.5). Suppose that (2.4) holds. For a sequence of nonnegative reals $(\lambda_Q)_{Q\in\mathcal{D}}$ we set

$$f(x) = \sup_{Q\in\mathcal{D}} \lambda_Q \, \chi_Q(x), \qquad x \in \mathbf{R}^n.$$

Obviously,

$$T_{K_{\mathcal{D}}}[f\,d\sigma](x) \ge \sum_{Q\in\mathcal{D}} K(Q) \, \lambda_Q \, \sigma(Q) \, \chi_Q(x).$$

Similarly, letting

$$g(x) = \sum_{Q\in\mathcal{D}} \lambda_Q \chi_Q(x), \qquad x \in \mathbf{R}^n,$$

we obtain

$$\begin{aligned}
T_{K_{\mathcal{D}}}[g\,d\sigma](x) &\ge \sum_{Q\in\mathcal{D}} K(Q) \, \chi_Q(x) \int_Q \left[\sum_{Q'\supset Q} \lambda_{Q'} \, \chi_{Q'}(y)\right] d\sigma(y) \\
&\ge \sum_{Q'\in\mathcal{D}} \lambda_{Q'} \, \sigma(Q') \overline{K}(Q') \, \chi_{Q'}(x).
\end{aligned}$$

Hence, the following two estimates hold:

$$\begin{aligned}
\left\| \sum_{Q\in\mathcal{D}} K(Q) \, \lambda_Q \, \sigma(Q) \, \chi_Q \right\|_{L^q(d\mu)} &\le C \left\| \sup_{Q\in\mathcal{D}} \lambda_Q \chi_Q \right\|_{L^p(d\sigma)}, \\
\left\| \sum_{Q\in\mathcal{D}} \overline{K}(Q) \, \lambda_Q \, \sigma(Q) \, \chi_Q \right\|_{L^q(d\mu)} &\le C \left\| \sum_{Q\in\mathcal{D}} \lambda_Q \chi_Q \right\|_{L^p(d\sigma)}.
\end{aligned} \tag{2.10}$$

Next, we rewrite the above two inequalities in terms of the discrete Littlewood–Paley spaces $\mathbf{f}^{0\,q}_{p\,\nu}$, $0 < p < +\infty$, $0 < q \le +\infty$, i.e., spaces which are formed by the collection of all sequences $(s_Q)_{Q\in\mathcal{D}}$ such that

$$\|(s_Q)_Q\|_{\mathbf{f}^{0\,q}_{p\,\nu}} = \left\| \left(\sum_{Q\in\mathcal{D}_\mu} (|s_Q| \nu(Q)^{-\frac{1}{2}} \chi_Q)^q \right)^{\frac{1}{q}} \right\|_{L^p(d\nu)} < +\infty,$$

where $\mathcal{D}_\nu$ is the set of dyadic cubes $Q \in \mathcal{D}$ such that $\nu(Q) \neq 0$. (See [**FrJ**], [**V1**].)

Replacing λ_Q by λ_Q^s, where $sq > 1$, we obtain

$$\left\| \sum_{Q\in\mathcal{D}} K(Q)\, \lambda_Q^s\, \sigma(Q)\, \chi_Q \right\|_{L^q(d\mu)} = \left\| \left(K(Q)^{\frac{1}{s}} \lambda_Q \sigma(Q)^{\frac{1}{s}} \mu(Q)^{\frac{1}{2}} \right)_{Q\in\mathcal{D}} \right\|^s_{\mathbf{f}^{0\,s}_{qs\,\mu}},$$

$$\left\| \sum_{Q\in\mathcal{D}} \overline{K}(Q)\, \lambda_Q^s\, \sigma(Q)\, \chi_Q \right\|_{L^q(d\mu)} = \left\| \left(\overline{K}(Q)^{\frac{1}{s}} \lambda_Q \sigma(Q)^{\frac{1}{s}} \chi_Q \right)_{Q\in\mathcal{D}} \right\|_{\mathbf{f}^{0\,s}_{qs\,\mu}},$$

and

$$\| \sup_{Q\in\mathcal{D}} \lambda_Q^s \chi_Q \|_{L^p(d\sigma)} = \| (\lambda_Q \sigma(Q)^{\frac{1}{2}}) \|^s_{\mathbf{f}^{0\,\infty}_{ps\,\sigma}},$$

$$\| \sum_{Q\in\mathcal{D}} \lambda_Q^s \chi_Q \|_{L^p(d\sigma)} = \| (\lambda_Q \sigma(Q)^{\frac{1}{2}}) \|^s_{\mathbf{f}^{0\,s}_{ps\,\sigma}}.$$

Consequently, the estimates in (2.10) are equivalent to

$$\left\| \left(K(Q)^{\frac{1}{s}} \lambda_Q \sigma(Q)^{\frac{1}{s}-\frac{1}{2}} \mu(Q)^{\frac{1}{2}} \right)_{Q\in\mathcal{D}} \right\|_{\mathbf{f}^{0\,s}_{qs\,\mu}} \leq C \left\| (\lambda_Q)_{Q\in\mathcal{D}} \right\|_{\mathbf{f}^{0\,\infty}_{ps\,\sigma}},$$

$$\left\| \left(\overline{K}(Q)^{\frac{1}{s}} \lambda_Q \sigma(Q)^{\frac{1}{s}} \chi_Q \right)_{Q\in\mathcal{D}} \right\|_{\mathbf{f}^{0\,s}_{qs\,\mu}} \leq C \left\| (\lambda_Q)_{Q\in\mathcal{D}} \right\|_{\mathbf{f}^{0\,s}_{ps\,\sigma}}.$$

It is shown in [**FrJ**], Corollary 8.3, that if $1 \leq p_0, q_0 < +\infty$, $1 \leq p_1, q_1 \leq +\infty$, $0 < \theta < 1$, $1/p = (1-\theta)/p_0 + \theta/p_1$ and $1/q = (1-\theta)/q_0 + \theta/q_1$, then the complex interpolation space $[\mathbf{f}^{0\,q_0}_{p_0\,dx}, \mathbf{f}^{0\,q_1}_{p_1\,dx}]_\theta \simeq \mathbf{f}^{0\,q}_{p\,dx}$, where dx is Lebesgue measure on $\mathbf{R}^n$. The main tool in their proof is the fact that the dyadic Hardy-Littlewood maximal operator (with respect to dx) is bounded in $L^p(dx)$, provided $p > 1$. Since the dyadic Hardy-Littlewood maximal operator with respect to an arbitrary nonnegative measure $d\mu$ is also bounded in $L^p(d\mu)$, it follows that $[\mathbf{f}^{0\,q_0}_{p_0\,\sigma}, \mathbf{f}^{0\,q_1}_{p_1\,\sigma}]_\theta \simeq \mathbf{f}^{0\,q}_{p\,\sigma}$. (See also [**CoV**].)

Using interpolation for analytic families of operators acting on the pair of normed spaces $\mathbf{f}^{0\,\infty}_{ps\,\sigma}$ and $\mathbf{f}^{0\,s}_{ps\,\sigma}$ (see, e.g., [**RS**], Theorem IX.20) we deduce:

$$\left\| \left(K(Q)^{\frac{1}{s}(1-\frac{1}{p})} \overline{K}(Q)^{\frac{1}{sp}} \lambda_Q \sigma(Q)^{\frac{1}{s}-\frac{1}{2}} \mu(Q)^{\frac{1}{2}} \right)_{Q\in\mathcal{D}} \right\|_{\mathbf{f}^{0\,s}_{qs\,\mu}} \leq C \left\| (\lambda_Q)_{Q\in\mathcal{D}} \right\|_{\mathbf{f}^{0\,ps}_{ps\,\sigma}},$$

and consequently

$$\left\| \sum_{Q\in\mathcal{D}} [K(Q)]^{1-\frac{1}{p}}\, [\overline{K}(Q)]^{\frac{1}{p}}\, \lambda_Q\, \sigma(Q)\, \chi_Q \right\|_{L^q(d\mu)}$$

$$\leq C \left\| \left(\sum_{Q\in\mathcal{D}} \lambda_Q^p \chi_Q \right)^{\frac{1}{p}} \right\|_{L^p(d\sigma)} = C \left[\sum_{Q\in\mathcal{D}} \lambda_Q^p\, \sigma(Q) \right]^{\frac{1}{p}}.$$

Now applying Theorem 3(c) in [**V1**] we conclude:

$$\sum_{Q\in\mathcal{D}} [\overline{K}(Q)]^{p'-1}\, K(Q)\, [\mu(Q)]^{p'-1}\, \sigma(Q)\, \chi_Q(x) = \mathcal{W}^d_{K,\sigma}[\mu](x) \in L^{\frac{q(p-1)}{p-q}}(d\mu).$$

This completes the proof of Theorem 2.1. □

3. Continuous trace inequalities

We recall that if $k : (0, +\infty) \to \mathbf{R}^+$ is a nonincreasing lower semicontinuous function, and σ is a nonnegative locally finite Borel measure on $\mathbf{R}^n$, we have defined in the Introduction the function

$$\overline{k}(r)(x) = \frac{1}{\sigma(B(x,r))} \int_0^r k(l)\, \sigma(B(x,l))\, \frac{dl}{l},$$

for $x \in \mathbf{R}^n$, $r > 0$. The continuous Wolff-type potential introduced in [**COV2**] is given by:

$$\mathcal{W}_{k,\sigma}[\mu](x) = \int_0^{+\infty} k(r)\sigma(B(x,r)) \left(\int_{B(x,r)} \overline{k}(r)(y) d\mu(y) \right)^{p'-1} \frac{dr}{r}. \tag{3.1}$$

We will assume that k and σ satisfy

$$\sup_{y \in B(x,r)} \overline{k}(r)(y) \leq A \inf_{y \in B(x,r)} \overline{k}(r)(y), \tag{3.2}$$

where A does not depend on $x \in \mathbf{R}^n$ and $r > 0$. In this case we will say that the pair (k, σ) has the logarithmic bounded oscillation property, or simply write $(k, \sigma) \in$ LBO.

If $(k, \sigma) \in$ LBO, the Wolff-type potential can be defined in an equivalent form given by

$$\mathcal{W}_{k,\sigma}[\mu](x) = \int_0^{+\infty} k(r)\sigma(B(x,r))\overline{k}(r)(x)^{p'-1}\mu(B(x,r))^{p'-1}\frac{dr}{r}.$$

If μ and σ are nonnegative Borel measures on $\mathbf{R}^n$, and $1 < p < +\infty$, the energy of μ associated with k and σ is given by

$$\mathcal{E}_{k,\sigma}[\mu] = \int_{\mathbf{R}^n} (T_k[\mu](x))^{p'}\, d\sigma(x). \tag{3.3}$$

We first observe that by setting $K(Q) = k(r_Q)$, we can associate to the radial kernel k a dyadic kernel $K_{\mathcal{D}}$, and the corresponding integral operator $T_{K_{\mathcal{D}}}$. The following proposition gives a relationship between the continuous energy and a supremum of the dyadic energy over shifts $\mathcal{D} + z$, $z \in \mathbf{R}^n$, where the shifted dyadic Wolff type potential is defined by

$$\mathcal{W}_{K,\sigma}^{\mathcal{D}+z}[\mu](x) = \sum_{Q \in \mathcal{D}} K(Q+z)\sigma(Q+z) \times \left(\int_{Q+z} \overline{K}(Q+z)(y) d\mu(y) \right)^{p'-1} \chi_{Q+z}(x), \quad x \in \mathbf{R}^n.$$

Proposition 3.1. *Let $k : (0, +\infty) \to \mathbf{R}^+$ be a nonincreasing lower semicontinuous function. Let $1 < p < +\infty$, and let σ be a nonnegative locally finite Borel measure on $\mathbf{R}^n$. Suppose that σ satisfies a doubling condition and that $(k, \sigma) \in$*LBO*. If $K(Q) = k(r_Q)$, then for any nonnegative Borel measure μ on $\mathbf{R}^n$,*

$$\mathcal{E}_{k,\sigma}[\mu] \simeq \sup_{z \in \mathbf{R}^n} \int_{\mathbf{R}^n} \mathcal{W}_{K,\sigma}^{\mathcal{D}+z}[\mu](x)\, d\mu(x), \tag{3.4}$$

with constants of equivalence that may depend on k and σ, but not on μ.

Proof of Proposition 3.1. In [**COV2**], Proposition 3.7, it is proved that there exist constants c, $C > 0$ such that for any $x \in \mathbf{R}^n$,

$$\sum_{Q\in\mathcal{D}} k(cr_Q)\sigma(Q)\chi_Q(x)\left(\int_Q \overline{K}(Q)(y)d\mu(y)\right)^{p'-1} \le C\,\mathcal{W}_{k,\sigma}[\mu](x). \tag{3.5}$$

The constant c above appears because, if Q is a cube and $x \in Q$, then $Q \subset B(x, \frac{c}{2}r_Q)$. Since $\overline{K}$ satisfies a doubling condition, the constant c is not needed in $\overline{K}$.

We also have (Lemma 3.9 in [**COV2**]) that for any $c > 0$,

$$\sum_{Q\in\mathcal{D}} k(cr_Q)\sigma(Q)\overline{k}(r_Q)^{p'-1}\mu(Q)^{p'} \simeq \sum_{Q\in\mathcal{D}} k(r_Q)\sigma(Q)\overline{k}(r_Q)^{p'-1}\mu(Q)^{p'}, \tag{3.6}$$

with constants that do not depend on μ. Consequently, (3.5) and (3.6) give that if we write $k_c(r) = k(cr)$, and $K_c(Q) = k_c(r_Q)$, then

$$\begin{aligned}\mathcal{E}_{k,\sigma}[\mu] &\simeq \int_{\mathbf{R}^n} \mathcal{W}_{k,\sigma}[\mu](x)d\mu(x)\\ &\ge C\sup_{z\in\mathbf{R}^n}\sum_{Q\in\mathcal{D}} K_c(Q+z)\sigma(Q+z)\overline{K_c}(Q+z)^{p'-1}\mu(Q+z)^{p'-1}\\ &\simeq \sup_{z\in\mathbf{R}^n}\int_{\mathbf{R}^n}\mathcal{W}^{\mathcal{D}+z}_{K,\sigma}[\mu](x)d\mu(x).\end{aligned}$$

On the other hand, the estimate obtained in [**COV2**], page 870, together with (3.6), gives

$$\mathcal{E}_{k,\sigma}[\mu] = \int_{\mathbf{R}^n} T_k[\mu](x)^{p'}d\sigma(x) \le C\sup_{z\in\mathbf{R}^n}\sum_{Q\in\mathcal{D}} K(Q+z)\sigma(Q+z)\overline{K}(Q+z)^{p'-1}\mu(Q+z)^{p'}.$$

□

Our next lemma shows that if the trace inequality holds for T_k, and $0 < q < +\infty$, $1 < p < +\infty$, then it is also valid for any operator T_{k_c} associated with the kernel $k_c(r) = k(cr)$, $c > 0$.

Lemma 3.2. *Let $k : (0, +\infty) \to \mathbf{R}^+$ be a nonincreasing lower semicontinuous function, and let $k_c(x) = k(cx)$, where $c > 0$. Let $1 < p < +\infty$ and $0 < q < \infty$. Suppose that σ and μ are nonnegative Borel measures on $\mathbf{R}^n$, and σ is a doubling measure. Then the following statements are equivalent:*

(i) The inequality

$$\int_{\mathbf{R}^n} |T_k[f d\sigma]|^q\, d\mu \le C_1\,\|f\|^q_{L^p(d\sigma)} \tag{3.7}$$

holds for all $f \in L^p(d\sigma)$.

(ii) The inequality

$$\int_{\mathbf{R}^n} |T_{k_c}[f d\sigma]|^q\, d\mu \le C_2\,\|f\|^q_{L^p(d\sigma)} \tag{3.8}$$

holds for all $f \in L^p(d\sigma)$.

Moreover, the least constants C_i, $i = 1, 2$, *in the above inequalities are equivalent, and the constants of equivalence depend only on* k, n, q, p, c, *and the doubling constant of* σ.

Proof of Lemma 3.2. Since k is nonincreasing it suffices to prove that (3.7) implies (3.8) for c small enough. Without loss of generality we may assume that $f \geq 0$ and $c = \frac{1}{2}$. Denote by M^σ the centered maximal function with respect to σ on $\mathbf{R}^n$ defined by

$$M^\sigma f(x) = \sup_{r>0} \frac{1}{\sigma(B(x,r))} \int_{B(x,r)} |f|\, d\sigma.$$

We then have

$$\begin{aligned} T_k[(M^\sigma f)d\sigma](x) &\geq \sum_{l\in\mathbb{Z}} \int_{2^{l-2}<|x-y|\leq 2^{l-1}} k(x-y) M^\sigma f(y)\, d\sigma(y) \\ &\geq \sum_{l\in\mathbb{Z}} k(2^{l-1}) \int_{2^{l-2}<|x-y|\leq 2^{l-1}} M^\sigma f(y)\, d\sigma(y). \end{aligned}$$

Notice that, for $y \in B(x, 2^{l-1})$ and $r = 2^{l+2}$,

$$B(x, 2^{l+1}) \subset B(y, r) \subset B(x, 2^{l+3}).$$

Hence, for $d\nu = f\, d\sigma$, we have

$$M^\sigma f(y) \geq \frac{1}{\sigma(B(y,r))} \int_{B(x,2^{l+1})} f\, d\sigma \geq C\, \frac{1}{\sigma(B(x,2^{l+3}))}\, \nu(B(x,2^{l+1})).$$

Since σ is a doubling measure, it follows (see [**COV2**], p. 874) that

$$\sigma(\{y : 2^{l-2} < |x-y| \leq 2^{l-1}\}) \simeq \sigma(B(x,2^{l+3})).$$

Thus,

$$T_k[(M^\sigma f)d\sigma](x) \geq C \sum_{l\in\mathbb{Z}} k(2^{l-1})\, \nu(B(x,2^{l+1})).$$

It is well-known (see, e.g., [**F**]) that M^σ is a bounded operator on $L^p(d\sigma)$, $p > 1$. Thus, it follows that (3.7) implies

$$\int_{\mathbf{R}^n} \left| \sum_{l\in\mathbb{Z}} k(2^{l-1}) \int_{B(x,\, 2^{l+1})} f\, d\sigma \right|^q d\mu \leq C\, \|f\|^q_{L^p(d\sigma)}. \tag{3.9}$$

It remains to check that in its turn (3.9) implies (3.8). Indeed, this follows from the estimate

$$T_{k_{\frac{1}{2}}}[f d\sigma](x) = \sum_{l\in\mathbb{Z}} \int_{2^l<|x-y|\leq 2^{l+1}} k_{\frac{1}{2}}(x-y)\, f(y)\, d\sigma(y) \leq \sum_{l\in\mathbb{Z}} k_{\frac{1}{2}}(2^l)\, \nu(B(x,2^{l+1})).$$

The proof of Lemma 3.2 is complete. □

Theorem 3.3. *Let $K : \mathcal{D} \to \mathbf{R}^+$, $0 < q < p < +\infty$, and $1 < p < +\infty$. Let μ and σ be nonnegative Borel measures on $\mathbf{R}^n$. Suppose that $(k, \sigma) \in$ (LBO), and σ satisfies a doubling condition. Then the trace inequality*

$$\int_{\mathbf{R}^n} |\, T_k[f d\sigma]\,|^q \, d\mu \le C \, \|f\|^q_{L^p(d\sigma)}, \qquad f \in L^p(d\sigma), \tag{3.10}$$

holds if and only if $\int_{\mathbf{R}^n} (\mathcal{W}_{k,\sigma}[\mu])^{\frac{q(p-1)}{p-q}} d\mu < +\infty$.

Proof of Theorem 3.3. The proof of the sufficiency is immediate from the following lemma, which is a continuous version of Lemma 2.2.

Lemma 3.4. *Under the same hypotheses as in Theorem 3.3, if*

$$d\mu_1 = \frac{d\mu}{(\mathcal{W}_{k,\sigma}[\mu])^{p-1}},$$

then

$$\int_{\mathbf{R}^n} |T_k[f d\sigma]|^p(x) \, d\mu_1(x) \le C \, \|f\|^p_{L^p(d\sigma)}. \tag{3.11}$$

Proof of Lemma 3.4. Duality and Lemma 3.2 show that (3.11) holds if and only if

$$\mathcal{E}_{k_c,\sigma}[g d\mu_1] \le C \|g\|^{p'}_{L^{p'}(d\mu_1)},$$

where $k_c(r) = k(cr)$, and $c > 0$ is the constant which appears in (3.5). Proposition 3.1 applied to $g\mu_1$ and k_c gives

$$\mathcal{E}_{k_c,\sigma}[g\mu_1] \simeq \sup_{z\in\mathbf{R}^n} \int_{\mathbf{R}^n} \mathcal{W}^{\mathcal{D}+z}_{K_c,\sigma}[g\mu_1](x)\, g(x) d\mu_1(x) \simeq \sup_{z\in\mathbf{R}^n} \mathcal{E}^{\mathcal{D}+z}_{K_c,\sigma}[g\mu_1].$$

Consequently, it remains to show that

$$\sup_{z\in\mathbf{R}^n} \int_{\mathbf{R}^n} T_{K_c\mathcal{D}+z}[f d\sigma]^p(x) \, d\mu_1(x) \le C \|f\|^p_{L^p(d\sigma)}.$$

But (3.6) yields that, for each $z \in \mathbf{R}^n$, $d\mu_1(x) \le C \frac{d\mu(x)}{\mathcal{W}^{\mathcal{D}+z}_{K_c,\sigma}[\mu](x)^{p-1}}$, with constant that does not depend on z. Applying Lemma 2.3, we complete the proof of Lemma 3.4. □

Now the proof of the sufficiency part in Theorem 3.3 follows the same argument as in the case of the discrete trace inequality.

To establish the necessity part, we start with a lemma.

Lemma 3.5. *Assume that (3.10) holds. Then there exist $c > 0$ which depends only on the dimension, and $C > 0$ such that, for any sequence (λ_Q) of nonnegative reals, and any $z \in \mathbf{R}^n$,*

$$\left\| \sum_{Q\in\mathcal{D}} [K_c(Q+z)]^{1-\frac{1}{p}} [\overline{K_c}(Q+z)]^{\frac{1}{p}} \lambda_Q \, \sigma(Q+z)\, \chi_{Q+z} \right\|_{L^q(d\mu)}$$

$$\le C \left[\sum_{Q\in\mathcal{D}} \lambda_Q^p \sigma(Q+z) \right]^{\frac{1}{p}}. \tag{3.12}$$

Proof of Lemma 3.5. If $(\lambda_Q)_Q$ is a sequence of nonnegative reals, we define $f(x) = \sup_{Q\in\mathcal{D}} \lambda_Q \chi_Q(x)$. For any $i \in \mathbf{Z}$, we denote by Q_i the unique cube in $\mathcal{D}$ suich that $x \in Q_i$ and $|Q_i| = 2^{in}$. We then have

$$T_k f(x) = \int_{\mathbf{R}^n} k(|x-y|) \sup_{Q\in\mathcal{D}} \lambda_Q \chi_Q(y) d\sigma(y) \geq$$
$$\sum_{i\in\mathbf{Z}} \lambda_{Q_{i+1}} \int_{Q_{i+1}\setminus Q_i} k(|x-y|)\chi_{Q_{i+1}}(y) d\sigma(y) = \sum_{i\in\mathbf{Z}} \lambda_{Q_{i+1}} \int_{Q_{i+1}\setminus Q_i} k(|x-y|) d\sigma(y).$$

Next, there exists $c > 0$ depending only on n, such that, for any $y \in Q_{i+1}$, $|x-y| \leq c2^{i+1} = cr_{Q_{i+1}}$. Since σ satisfies a doubling condition, we also have that $\sigma(Q_{i+1}\setminus Q_i) \simeq \sigma(Q_{i+1})$. Altogether, we deduce that the above sum is bounded from below by

$$C\sum_{i\in\mathbf{Z}} \lambda_{Q_{i+1}} k_c(r_{Q_{i+1}})\sigma(Q_{i+1})\chi_{Q_{i+1}}(x) = \sum_{Q\in\mathcal{D}} \lambda_Q K_c(Q)\sigma(Q)\chi_Q(x).$$

On the other hand, let $g(x) = \sum_{Q\in\mathcal{D}} \lambda_Q \chi_Q(x)$. We then have

$$T_k g(x) = \int_{\mathbf{R}^n} k(|x-y|) \sum_{Q\in\mathcal{D}} \lambda_Q \chi_Q(y) d\sigma(y) \geq \sum_{x\in Q} \lambda_Q \int_Q k(|x-y|) d\sigma(y).$$

If $x \in Q$, we denote by $2^{-l}Q$, $l \geq 0$, the unique cube in $\mathcal{D}$ satisfying $x \in 2^{-l}Q$ and $r_{2^{-l}Q} = 2^{-l} r_Q$. Then

$$\int_Q k(|x-y|) d\sigma(y) = \sum_{l\geq 0} \int_{2^{-l}Q\setminus 2^{-l-1}Q} k(|x-y|) d\sigma(y) \geq$$
$$\sum_{l\geq 0} k_c(2^{-l}r_Q)\sigma(2^{-l}Q \setminus 2^{-l-1}Q) \simeq \sum_{l\geq 0} k_c(2^{-l}r_Q)\sigma(2^{-l}Q) \simeq \overline{K_c}(Q)\sigma(Q).$$

Consequently,

$$T_k g(x) \geq C \sum_{Q\in\mathcal{D}} \lambda_Q \sigma(Q)\overline{K_c}(Q)\chi_Q(x).$$

We then have:

$$\left\| \sum_{Q\in\mathcal{D}} \lambda_Q K_c(Q)\, \sigma(Q)\chi_Q \right\|_{L^q(d\mu)} \leq C \left\| \sup_{Q\in\mathcal{D}} \lambda_Q \chi_Q \right\|_{L^p(d\sigma)},$$
$$\left\| \sum_{Q\in\mathcal{D}} \lambda_Q \overline{K_c}(Q)\, \sigma(Q)\chi_Q \right\|_{L^q(d\mu)} \leq C \left\| \sum_{Q\in\mathcal{D}} \lambda_Q \chi_Q \right\|_{L^p(d\sigma)}.$$

We can now employ the interpolation argument used in the proof of the necessity for the discrete trace inequality to obtain

$$\left\| \sum_{Q\in\mathcal{D}} [K_c(Q)]^{1-\frac{1}{p}} [\overline{K_c}(Q)]^{\frac{1}{p}} \lambda_Q \sigma(Q)\chi_Q \right\|_{L^q(d\mu)} \leq C \left[\sum_{Q\in\mathcal{D}} \lambda_Q^p \sigma(Q) \right]^{\frac{1}{p}}.$$

Applying the same argument to a shifted dyadic lattice $\mathcal{D} + z$, $z \in \mathbf{R}^n$, we obtain a similar inequality with a constant C which does not depend on z. □

In order to follow with the proof of the necessity, we recall that by Theorem 3 (c) in [**V1**], estimate (3.12) is equivalent to the condition $\mathcal{W}^{\mathcal{D}+z}_{K_c,\sigma}[\mu] \in L^{\frac{q(p-1)}{p-q}}(d\mu)$, uniformly in z, where

$$\mathcal{W}^{\mathcal{D}+z}_{K_c,\sigma}[\mu](x) = \sum_{Q\in\mathcal{D}} [\overline{K_c}(Q+z)]^{p'-1}\, K_c(Q+z)\, [\mu(Q+z)]^{p'-1}\, \sigma(Q+z)\, \chi_{Q+z}(x).$$

Our next goal is to show that the proof of the above result can be modified to obtain that if $B_j = B(0, 2^{j+j_0})$ and $x \in B_j$ (here j_0 is fixed so that $2^{j_0} > 2\sqrt{n}+1$), then (3.13)

$$\sum_{Q\in\mathcal{D}} \left[\frac{1}{|B_j|}\int_{B_j} \Big([\overline{K}_c(Q+z)]^{p'-1}\, K_c(Q+z)\, [\mu(Q+z)]^{p'-1}\, \sigma(Q+z)\, \chi_{Q+z}(x)\Big)^{\frac{p-1}{ps-1}} dz\right]^{\frac{ps-1}{p-1}}$$

belongs to $L^{\frac{q(p-1)}{p-q}}(d\mu)$, where $s>1$ satisfies $sq>1$.

Indeed, if $s>1$ is chosen so that $sq>1$, (3.12) can be rewritten as

$$\sup_{z\in\mathbf{R}^n} \left\| \sum_{Q\in\mathcal{D}} [K_c(Q+z)]^{(1-\frac{1}{p})\frac{1}{s}} [\overline{K_c}(Q+z)]^{\frac{1}{ps}} \lambda_Q \sigma(Q+z)^{\frac{1}{sp'}} \chi_{Q+z} \right\|_{L^{qs}(l^{ps})(d\mu)} \le C\|(\lambda_Q)\|_{l^{ps}}.$$

We then have, arguing as in [**V1**], p. 545, that the above inequality is equivalent to

$$\sup_{z\in\mathbf{R}^n} \left(\sum_{Q\in\mathcal{D}} \Big([K_c(Q+z)]^{(1-\frac{1}{p})\frac{1}{s}} [\overline{K_c}(Q+z)]^{\frac{1}{ps}} \sigma(Q+z)^{\frac{1}{sp'}}\Big)^{\frac{ps}{ps-1}} \left| \int_{Q+z} g_Q d\mu \right|^{\frac{ps}{ps-1}} \right)^{\frac{ps-1}{ps}}$$
$$\le C\, ||g||_{L^{\frac{qs}{qs-1}}(l^{s'})(d\mu)},$$

for all $g=(g_Q) \in L^{\frac{qs}{qs-1}}(l^{s'})(d\mu)$. Here $L^s(l^r)(d\mu)$ denotes the mixed-norm Lebesgue space of vector-valued functions $g=(g_Q)$ equipped with the quasi-norm

$$||g||_{L^s(l^r)(d\mu)} = \left[\int_{\mathbf{R}^n} \Big(\sum_Q |g_Q|^r\Big)^{\frac{s}{r}} d\mu\right]^{\frac{1}{s}}.$$

Setting $g_Q = \psi_Q^{\frac{ps-1}{ps}}$, $\psi_Q \ge 0$, and using Hölder's inequality together with the dyadic vector-valued Fefferman-Stein maximal theorem, it is easy to see that the last estimate is equivalent to

$$\sup_{z\in\mathbf{R}^n} \sum_{Q\in\mathcal{D}} \Big([K_c(Q+z)]^{(1-\frac{1}{p})\frac{1}{s}} [\overline{K_c}(Q+z)]^{\frac{1}{ps}} \sigma(Q+z)^{\frac{1}{sp'}} \mu(Q+z)\Big)^{\frac{ps}{ps-1}}$$
$$\times \frac{1}{\mu(Q+z)} \int_{Q+z} \psi_Q d\mu \le C\, ||\psi||_{L^{\frac{ps-1}{qs-1}\frac{q}{p}}(l^{\frac{ps-1}{s-1}\frac{1}{p}})(d\mu)},$$

for all $\psi = (\psi_Q)_Q \in L_\mu^{\frac{ps-1}{qs-1}\frac{q}{p}}(l^{\frac{ps-1}{s-1}\frac{1}{p}})$. (See the proof of Theorem 3(c) in [**V1**] for details.) Averaging over $z \in B_j$ gives

$$\sum_{Q\in\mathcal{D}} \frac{1}{|B_j|}\int_{B_j}\Big([K_c(Q+z)]^{(1-\frac{1}{p})\frac{1}{s}}[\overline{K_c}(Q+z)]^{\frac{1}{ps}}\sigma(Q+z)^{\frac{1}{sp'}}\mu(Q+z)\Big)^{\frac{ps}{ps-1}}\times$$
$$\frac{1}{\mu(Q+z)}\int_{Q+z}\psi_Q\,d\mu\,dz \le C\,||\psi||_{L^{\frac{ps-1}{qs-1}\frac{q}{p}}(l^{\frac{ps-1}{s-1}\frac{1}{p}})(d\mu)}.$$

The preceding estimate by duality yields (3.13).

Let $c>0$ be the constant in Lemma 3.5. As was shown in [**COV2**], p. 877, for the truncated Wolff type potential associated with k_{2c} defined by

$$\mathcal{W}^R_{k_{2c},\sigma}[\mu](x) = \int_0^R k_{2c}(r)\sigma(B(x,r))\overline{k_{2c}}(r)^{p'-1}(x)\mu(B(x,r))^{p'-1}\frac{dr}{r},$$

it follows that, for every $l\le j$ and $x\in B_j = B(0,2^j)$,

$$\mathcal{W}^{2^j}_{k_{2c},\sigma}[\mu](x) \le C\sum_{l\le j}\sigma(B(x,2^l))k_{2c}(2^{l-1})\overline{k_{2c}}(2^l)(x)^{p'-1}\mu(B(x,2^l))^{p'-1}.$$

Next, for any $l\le j$,

$$[\overline{k}_{2c}(2^l)]^{p'-1}\,k_{2c}(2^{l-1})\,[\mu(B(x,2^l))]^{p'-1}\,\sigma(B(x,2^l)) \preceq$$
$$\left(\frac{1}{|B_j|}\int_{B_j}\Big([\overline{K}_c(Q_l+z)]^{p'-1}\,K_c(Q_l+z)\,[\mu(Q_l+z)]^{p'-1}\,\sigma(Q_l+z)\,\chi_{Q_l+z}(x)\Big)^{\frac{p-1}{ps-1}}\,dz\right)^{\frac{ps-1}{p-1}},$$

where Q_l+z is the unique cube of side length $r(Q+z)=2^{l+1}$ that contains x. This follows as in the proof of Theorem 3.13 in [**COV2**] by noticing that all sums over the generation of cubes $Q+z$ such that $r_{Q+z}=2^{l+1}$ on p. 877 in [**COV2**] actually contain only one term. Hence,

$$\int_{B_j}\left(\sum_{l\le j}[\overline{k}_{2c}(2^l)]^{p'-1}\,k_{2c}(2^{l-1})\,[\mu(B(x,2^l))]^{p'-1}\,\sigma(B(x,2^l))\right)^{\frac{q(p-1)}{p-q}}d\mu \le C < +\infty.$$

Letting $j\to+\infty$, we obtain $\int_{\mathbf{R}^n}(\mathcal{W}_{k_{2c},\sigma}[\mu])^{\frac{q(p-1)}{p-q}}d\mu<+\infty$.

Now using the same argument with the kernel $k_{\frac{1}{2c}}$ in place of k and applying Lemma 3.2, we complete the proof of Theorem 3.3. □

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